

FINAL EXPLANATION: CORE MATHEMATICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: CORE MATHEMATICS GRADE 10: SUB-STRAND 2.1 SIMILARITY AND ENLARGEMENT

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

This is your Final Explanation for the unit. Think back to the Kisumu school plan labelled Scale 1 : 200 and the two water tanks, one of which holds 8 times as much water as the other. Answer every section in your own words. Use the evidence you collected in Lessons 1 to 8, including your measurements, tables and models. Show all your working for calculations and write full sentences when asked to explain. Use the words linear scale factor (L.S.F), area scale factor (A.S.F), volume scale factor (V.S.F) and centre of enlargement correctly. Your teacher will mark your work against the rubric.

Section 1: What makes figures similar, and what does Scale 1 : 200 mean?

The Kisumu school plan is labelled Scale 1 : 200. A classroom block that is 10 m wide appears on the plan with a certain width. (a) State the two conditions that must be true for two figures to be similar. (b) Calculate the width of the classroom block on the plan in centimetres, and write the L.S.F as a fraction. (c) Explain why every window, door and corridor on the plan is still in the right proportion even though the whole school compound fits on one A3 sheet.

Write your answer here:

Section 2: Centre of enlargement and constructing images

A triangle ABC represents a classroom block. It is enlarged from a centre O with L.S.F $k = 2$ and then separately with $k = -2$. (a) Explain how you can find the centre of enlargement when you are given an object and its image. (b) Describe where the image is placed, and how it is oriented, for $k = 2$ and for $k = -2$. (c) The point A is at (2, 1) and the centre is the origin. Find the image of A for $k = 2$ and for $k = -2$ using the formula $\text{Image} = \text{Centre} + k \times (\text{Object} - \text{Centre})$.

Write your answer here:

Section 3: How area changes under enlargement

In Lesson 5 you compared similar rectangles and triangles. (a) A student predicts that doubling every length of a rectangle doubles its area. Use a rectangle 3 cm by 2 cm and its enlargement with L.S.F 2 to test this prediction. (b) State the rule connecting A.S.F and L.S.F and explain why it works. (c) The base of a 2 kg Unga Pembe flour packet has an area 4 times that of the 1 kg packet. What is the L.S.F between the two bases? (d) At Scale 1 : 200, how many times smaller is an area on the plan than the real area?

Write your answer here:

Section 4: Solving the water tank puzzle

The school has two water tanks of identical shape. The larger tank is twice as tall as the smaller tank. (a) State the rule connecting V.S.F and L.S.F. (b) Use the rule to work out how many times as much water the larger tank holds as the smaller tank, and explain why your answer is not 2. (c) Copy and complete this table for pairs of similar solids, then show the calculations for the last row.

L.S.F	A.S.F	V.S.F
2		
3		

(d) A sufuria has a capacity of 2 litres. A similar sufuria has an L.S.F of 3 compared with it. What is the capacity of the larger sufuria?

Write your answer here:

Section 5: Answering the driving question

Driving question: When you scale up a shape or solid, such as doubling the size of a water tank or enlarging a building plan, what exactly happens to its lengths, its area and its volume, and how can we use this to solve real problems in our community? (a) Write a paragraph that answers the driving question using the terms L.S.F, A.S.F and V.S.F and referring to the school plan and the water tanks. (b) Give one new real-life Kenyan example, different from the ones used in class, where you would use these scale factors, and show a calculation. (c) Complete this reflection: 'In Lesson 1 I used to think ... Now I know ...'

Write your answer here:

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Understanding of similarity and scale	States both conditions for similar figures and correctly calculates the plan width and L.S.F of 1/200, explaining clearly why proportions are preserved.	States both conditions and calculates the L.S.F correctly, with a partly complete explanation of proportion.	States only one condition or makes errors in converting units or finding the L.S.F.
Enlargement and centre of enlargement	Explains how to find the centre using ray lines, describes positive and negative k accurately, and calculates image coordinates correctly.	Describes the centre and the effect of k with minor gaps, and gets most coordinates correct.	Cannot describe how to find the centre or confuses positive and negative k.
Area and volume scale factors	Correctly applies $A.S.F = (L.S.F)^2$ and $V.S.F = (L.S.F)^3$ in all calculations and explains why each rule works.	Applies both rules correctly in most calculations with a limited explanation.	Treats area or volume as scaling by k, or makes frequent calculation errors.

Explanation and use of evidence	Connects the school plan and water tanks to the scale factor rules with clear, complete reasoning and accurate use of terms.	Connects the phenomenon to the rules with generally accurate terms but some reasoning left implicit.	Gives statements with little reasoning or little link to the phenomenon.
Real-life application and reflection	Gives a new, realistic Kenyan example with a correct calculation and a clear 'used to think / now I know' reflection.	Gives a suitable example with a mostly correct calculation and a brief reflection.	Example is unsuitable or the calculation is incorrect, and the reflection is missing or vague.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

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